

Apt Interview Assignment

Problem Statement

You are required to design and implement a system where clients automatically receive updates whenever data in the database changes. The system should not rely on frequent polling from clients. The challenge is to think about how updates can be efficiently propagated from the database to connected clients in real-time.

Requirements

- **Database Changes** – Assume you have a table called 'orders' with fields: id (int, primary key), customer_name (string), product_name (string), status (string: 'pending', 'shipped', 'delivered'), updated_at (timestamp). Any insert/update/delete on this table should trigger an update to clients.
- **Client Updates** – When something changes in the database, they should get notified automatically with the new data.
- **Tech Choices** – You may use any backend language (Node.js, Python, Java, etc.). Database can be MySQL/Postgres/MongoDB (your choice). You are free to use libraries or frameworks that help solve the problem.
- **Expected Deliverables** – A working backend service that listens for DB changes and pushes updates to clients. A simple client (CLI, browser, or script) that shows updates in real time. Documentation (README) explaining your approach, how to run your solution, and why you chose your method.

Evaluation Criteria

- Design Thinking – Did you consider scalability and efficiency?
- Correctness – Does the system send real-time updates correctly?
- Code Quality – Is the code clean, modular, and readable?
- Documentation – Is it easy for others to understand and run?

The goal is not just coding but also showing how you think about building real-time systems.